DESIGN COMPUTED · FREE GUIDE

Record it before it's gone: how a laser scan helped rebuild Notre-Dame's roof

The free guide to the reel on Notre-Dame's oak roof frame: what the 'forest' was, how a 2010 laser scan became a reference for the rebuild, how carpenters worked the new oak, and how to scan a place yourself with your phone.



The forest in the roof

Above the stone vaults of Notre-Dame de Paris sat a medieval oak roof frame so dense with beams that it was nicknamed 'the forest'. When the cathedral burned in April 2019, around two thirds of the roof was destroyed, together with the spire.

A roof frame is not decoration. Its triangles carry the weight of the roof and the wind loads down into the stone walls, so rebuilding it meant rebuilding a structure, not only a look.

A scan made nine years earlier

In 2010 the art historian Andrew Tallon, working with Paul Blaer, scanned Notre-Dame with a laser scanner. They moved the scanner to fifty positions and recorded over a billion points of data.

A laser scanner measures how long a beam takes to bounce back from a surface. Do that millions of times in every direction and you get a 'point cloud': a 3D map in which each dot has an exact position. After the fire, this mapping was used as a resource for the reconstruction.

Rebuilding the forest

Carpenters rebuilt the frame by hand from fresh oak, shaping the beams with axes in the medieval way, and the frame was reported finished in 2024. The cathedral reopened in December 2024, more than five years after the fire.

Built examples of 'record it first'

Notre-Dame de Paris	A 2010 laser scan of the interior later served as a reference for reconstruction.
Heritage surveys in general	Laser scanning and photogrammetry are used to keep exact records of monuments before conservation work or damage.
Your own city	Community projects use phone photos or free apps to capture buildings at risk.

Try it yourself: scan a small place with your phone

1. Pick a small object or corner with texture: a statue, a doorway, a stone wall.
2. Install a free photogrammetry or scanning app (many phone apps and the free desktop tool Meshroom work).
3. Walk around the subject taking 40 to 60 overlapping photos, keeping the light even and the subject sharp.
4. Load the photos in the app and let it compute a 3D model, a point cloud first and then a surface.
5. Rotate the model and look at what it records that your eyes missed: cracks, depths, small details. That is the value of a scan.

The rule to remember

Record what you care about before you need it: a measured record turns a loss into a rebuild.

Default answer	Rely on memory and old drawings.
Form-finder answer	Capture it with measurement while it is whole.
Check	Ask: if this disappeared tomorrow, what would we have to rebuild it from?

Go further

Andrew Tallon (Wikipedia)
https://en.wikipedia.org/wiki/Andrew_Tallon

Notre-Dame de Paris (Wikipedia)
https://en.wikipedia.org/wiki/Notre-Dame_de_Paris

NPR: carpenters rebuild Notre-Dame's roof
https://www.npr.org/2023/06/11/1179648233/notre-dame-paris-fire-rebuild-roof

Credits
Cover photo: Ibex73, CC BY-SA 4.0, Wikimedia Commons.
Facts: Wikipedia (Andrew Tallon), NPR, The Art Newspaper, Wood Central, AP reporting on the reopening.

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